MINI PROJECT REPORT

SALES DATA CLEANING AND ANALYSIS USING PYTHON

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1. Introduction

This project involved examining, cleaning and analyzing a sales dataset using Python in Google Colab. The original file contained inconsistent text, missing values, duplicate records, mixed data formats and values that needed correction. The purpose was to improve the quality of the dataset for basic sales analysis and visualization.

2. Objectives

The objectives were to load the supplied CSV file into Google Colab; inspect its structure and data quality; identify missing values and duplicate records; correct inconsistent text and incorrectly formatted values; convert suitable columns to appropriate data types; calculate a total order amount; create charts; and export the cleaned dataset as a CSV file.

3. Dataset Description

The dataset was supplied as Messy_data.csv. Initial inspection showed 185 records and 9 columns: OrderID, CustomerName, City, Product, Quantity, UnitPrice, OrderDate, OrderStatus and CustomerAge. These fields describe orders, customers, products, prices, dates and order statuses.

4. Tools Used

The project was completed in Google Colab using Python. The pandas library was used to load, inspect, clean and summarize the data. Matplotlib was used to create visualizations. The cleaned dataset was exported using pandas to_csv().

5. Data Inspection

The dataset was inspected with head(), shape, info() and isnull().sum(). The initial shape was 185 rows by 9 columns. The inspection showed missing values in several fields, including customer name, city, quantity, unit price, order date, order status and customer age. A duplicate check identified 5 duplicate rows.

6. Data Cleaning Procedures

A working copy of the original DataFrame was created so that the source data remained unchanged. Leading and trailing spaces were removed from text fields. City names and order statuses were standardized for consistent capitalization. Examples of inconsistent values were corrected, including quantity values such as 'three' and '2 pcs', the price value '18,500', the text value 'unknown', and the age value 'forty'. The age entry '29 years' was also converted to a numeric value. Suitable numeric columns were converted using pandas.to_numeric(), with values that could not be interpreted as numbers treated as missing. Order dates were converted to datetime format. Five duplicate rows were removed, leaving 180 records.

7. Missing Values and Outliers

Missing categorical values in CustomerName, City, Product and OrderStatus were labelled 'Unknown' where appropriate. Missing or invalid numeric values were retained as missing rather than inventing values. Invalid dates were converted to missing dates. Potentially suspicious values were checked using simple range rules: quantity values below 0 or above 100, unit prices below 0 or above 1,000,000, and customer ages below 0 or above 120 were treated as missing. These are screening rules used for this project, not proof that every flagged value was an error.

8. Derived Column and Analysis

A new column named TotalAmount was created by multiplying Quantity by UnitPrice for each record. This supports basic order-value analysis. Descriptive statistics were generated for numeric fields, and the cleaned data was reviewed to understand quantities, prices and customer ages. Records with missing quantity or unit price do not have a calculable TotalAmount.

9. Visualizations

Bar charts were created to show the number of orders by order status and by city. These charts provide a simple comparison of order categories and the locations represented in the dataset. The charts were generated with Matplotlib.

10. Results

The original dataset contained 185 rows and 9 columns. After removing 5 duplicate rows, 180 rows remained. The text and numeric formatting issues identified during inspection were addressed, relevant columns were converted to suitable types, and the TotalAmount field and two bar charts were produced. The cleaned data was saved as cleaned_sales_dataset.csv.

11. Conclusion

The project demonstrated how Python and pandas can be used to improve the quality of a messy sales dataset. The work included inspection, text standardization, correction of selected data-entry errors, type conversion, duplicate removal, missing-value handling, outlier screening, calculation of total order amounts and visualization. The cleaned CSV and Google Colab notebook provide the main practical outputs.

12. Files Submitted

The project submission includes the Google Colab notebook (.ipynb), the cleaned dataset (cleaned_sales_dataset.csv), and this written report. The original Messy_data.csv file was used as the input.

Appendix: Main Python Operations

The notebook contains the executable Python cells used for the project. The main operations included reading the CSV with pandas, inspecting the DataFrame, creating a working copy, standardizing text,

converting numeric and date columns, removing duplicates, screening unusual values, calculating TotalAmount, creating bar charts and exporting the cleaned CSV.